

KSP Datathon 2025

Intelligent Conversational AI for Crime Database

15-Day Sprint Plan · 6 July – 19 July 2025

Team · Gopika S · Abishek K S · Dharaneesh J · Vishal K

Stack: React · FastAPI · Supabase · Neo4j · LangChain · AI4Bharat

Phase 1 · Foundation	Phase 2 · Core Build	Phase 3 · Intelligence	Phase 4 · Polish & Demo
Jul 6–8	Jul 9–13	Jul 14–16	Jul 17–19
Gopika S Full Stack · Frontend	Abishek K S Full Stack · PM	Dharaneesh J Database Manager	Vishal K ML Engineer

Workflow Chart — Day-wise Member Tasks

Date	Gopika S	Abishek K S	Dharaneesh J	Vishal K
Foundation				
Jul 6 Project Setup & Schema	- Create React app (Vite) - Build chat UI shell - Setup GitHub pages	- Init FastAPI project - Create GitHub repo & invite team - Define API endpoint list	- Create Supabase project - Run full SQL schema - Setup storage buckets	- Setup Python ML environment - Install scikit-learn & transformers - Explore DBSCAN on dummy coords
Jul 7 Data Seeding & API Skeleton	- Chat message component - Voice input basic test - Route pages (chat, dashboard)	- Build /query & /cases endpoints - Integrate Supabase client - Auth middleware stub	- Seed 30 dummy cases - Seed suspects, items, recovery - Upload 10 sample photos	- Test pgvector extension - Generate FIR text embeddings - Store vectors in DB
Jul 8 First Integration	- Connect React → FastAPI - Display case list in UI - Basic Leaflet map component	- LLM first working test - Text-to-SQL basic prompt - End-to-end: question→SQL→result	- Test all cross-table joins - SHA-256 hash on evidence - Audit log test inserts	- MO similarity: top-3 cases - Cosine similarity via pgvector - Cross-case match logic draft
Core Build				
Jul 9 LLM + Conversational Core	- Multi-turn chat history UI - Loading states & error handling - Mobile responsive layout	- LangChain multi-turn context - Schema-aware text-to-SQL - Intent classifier (3 types)	- Neo4j Aura project setup - Graph schema: Person→Case nodes - First Cypher queries working	- DBSCAN on seeded case coords - Hotspot zones → GeoJSON - Prophet: monthly crime trend
Jul 10 Voice + Kannada Pipeline	- Whisper English speech→text - AI4Bharat Kannada STT test - Voice button in chat UI	- Translation endpoint pipeline - Kanglish mixed-input handling /voice-query API endpoint	- Neo4j: insert suspects+cases - Verify graph relationships - Basic network traversal queries	- Hotspot GeoJSON for frontend - Confidence score per answer - Risk scoring: repeat offender

Jul 11 Photo Evidence Pipeline	- Evidence photo gallery in chat - Suspect photo card component - Recovery items photo grid	- Upload endpoint → Supabase Storage /evidence/{case_id} API - Signed URL for secure access	- Test photo pipeline end-to-end - Verify SHA-256 chain of custody - Recovery items → persons link	- MO match → similarity % - Explainability: matched fields - Write ML output spec for API
Jul 12 Graph Viz + Hotspot Map	- Cytoscape.js criminal network - Leaflet hotspot cluster markers - Click node → case detail	/network/{suspect_id} API /hotspots GeoJSON API - Wire ML outputs to endpoints	- Neo4j: location nodes + edges - Cross-district case links - Performance test Neo4j queries	- Finalise DBSCAN output format - Prophet trend chart data - Early warning rising-area flag
Jul 13 RBAC + Audit + Buffer	- Login page with role selector - Role-based UI (3 levels) - Fix UI bugs from week 1	- RBAC middleware (FastAPI) - Audit log: log every query /audit-log admin endpoint	- Test RBAC row-level security - Verify audit captures all queries - Fix any schema issues	- Improve model on dummy data - Test all ML endpoints Postman - Document ML module API
Intelligence				
Jul 14 Cross-Case Linking	'Similar cases' panel in chat - Show matching scores + reasons - Highlight matched MO fields	- Proactive cross-case alert - LLM: explain match in language - Test: 'is this part of series?'	- Neo4j: SIMILAR_MO relationship - Query: same suspect+location - Seed 5 case series for demo	- Tune MO similarity threshold - Explainability: 3 key fields - Cross-district suspect match
Jul 15 Predictive Intelligence	- Trend chart (Chart.js) - Early warning badge on map - PDF export button	- LLM predictive questions - Explanation with source records - Wire Prophet into chat answers	- Seed seasonal pattern data - Verify trend date range queries - Index on date + location cols	- Prophet per-crime-type forecast - Early warning threshold logic - Top 3 predicted hotspots
Jul 16 Full Flow Test + Fixes	- End-to-end UI test all features - Fix display/render issues - PDF export with history working	- Full API integration test - Fix LangChain context bugs - Latency check: target <3s	- DB integrity check all tables - Load test: 10 concurrent queries - Backup Supabase before demo	- ML accuracy check all 3 models - Fix prediction edge cases - Document confidence meanings
Polish & Demo				
Jul 17 Demo Script + Dry Run 1	- Polish UI colours & spacing - Record 3-min demo screen flow - Fix any mobile layout issues	- Write demo script (exact queries) - Dry run 1: full team walkthrough - Prepare judge Q&A; doc	- Prepare 'wow' demo data - Keep-alive ping for Supabase - Ensure Neo4j graph is rich	- Prepare ML insight visuals - Verify model outputs in UI - Dry run: cross-case match live
Jul 18 Pitch Deck + Dry Run 2	- Final UI polish pass - Screenshots for pitch deck - Test on multiple browsers	- Build pitch deck (full slides) - Dry run 2: timed 5+3 min - Prepare 1-page summary	- Final evidence hashes check - Verify all photos load in demo - Document schema for submission	- Explainability panel demo moment - Write ML approach for deck - Full early warning alert test
Jul 19 Final Day — Submit & Present	- Production build React app - Deploy to Vercel / Netlify - Final UI smoke test live URL	- Deploy FastAPI to Railway/Render - Final API smoke test production - Submit project & deck	- Final Supabase + Neo4j check - Keep DB active during pitch - Ready for DB judge questions	- Final ML test on production - Backup static ML outputs ready - Present ML section in pitch

Detailed Day-by-Day Plan

Phase 1 · Foundation (Jul 6–8)

Jul 6 — Project Setup & Schema

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Create React app (Vite) • Build chat UI shell • Setup GitHub pages	• Init FastAPI project • Create GitHub repo & invite team • Define API endpoint list	• Create Supabase project • Run full SQL schema • Setup storage buckets	• Setup Python ML environment • Install scikit-learn & transformers • Explore DBSCAN on dummy coords

Jul 7 — Data Seeding & API Skeleton

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Chat message component • Voice input basic test • Route pages (chat, dashboard)	• Build /query & /cases endpoints • Integrate Supabase client • Auth middleware stub	• Seed 30 dummy cases • Seed suspects, items, recovery • Upload 10 sample photos	• Test pgvector extension • Generate FIR text embeddings • Store vectors in DB

Jul 8 — First Integration

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Connect React → FastAPI • Display case list in UI • Basic Leaflet map component	• LLM first working test • Text-to-SQL basic prompt • End-to-end: question→SQL→result	• Test all cross-table joins • SHA-256 hash on evidence • Audit log test inserts	• MO similarity: top-3 cases • Cosine similarity via pgvector • Cross-case match logic draft

Phase 2 · Core Build (Jul 9–13)

Jul 9 — LLM + Conversational Core

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Multi-turn chat history UI • Loading states & error handling • Mobile responsive layout	• LangChain multi-turn context • Schema-aware text-to-SQL • Intent classifier (3 types)	• Neo4j Aura project setup • Graph schema: Person→Case nodes • First Cypher queries working	• DBSCAN on seeded case coords • Hotspot zones → GeoJSON • Prophet: monthly crime trend

Jul 10 — Voice + Kannada Pipeline

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
- Whisper English speech→text - AI4Bharat Kannada STT test - Voice button in chat UI	- Translation endpoint pipeline - Kanglish mixed-input handling /voice-query API endpoint	- Neo4j: insert suspects+cases - Verify graph relationships - Basic network traversal queries	- Hotspot GeoJSON for frontend - Confidence score per answer - Risk scoring: repeat offender

Jul 11 — Photo Evidence Pipeline

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
- Evidence photo gallery in chat - Suspect photo card component - Recovery items photo grid	- Upload endpoint → Supabase Storage /evidence/{case_id} API - Signed URL for secure access	- Test photo pipeline end-to-end - Verify SHA-256 chain of custody - Recovery items → persons link	- MO match → similarity % - Explainability: matched fields - Write ML output spec for API

Jul 12 — Graph Viz + Hotspot Map

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
- Cytoscape.js criminal network - Leaflet hotspot cluster markers - Click node → case detail	/network/{suspect_id} API /hotspots GeoJSON API - Wire ML outputs to endpoints	- Neo4j: location nodes + edges - Cross-district case links - Performance test Neo4j queries	- Finalise DBSCAN output format - Prophet trend chart data - Early warning rising-area flag

Jul 13 — RBAC + Audit + Buffer

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
- Login page with role selector - Role-based UI (3 levels) - Fix UI bugs from week 1	- RBAC middleware (FastAPI) - Audit log: log every query /audit-log admin endpoint	- Test RBAC row-level security - Verify audit captures all queries - Fix any schema issues	- Improve model on dummy data - Test all ML endpoints Postman - Document ML module API

Phase 3 · Intelligence (Jul 14–16)

Jul 14 — Cross-Case Linking

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• 'Similar cases' panel in chat • Show matching scores + reasons • Highlight matched MO fields	• Proactive cross-case alert • LLM: explain match in language • Test: 'is this part of series?'	• Neo4j: SIMILAR_MO relationship • Query: same suspect+location • Seed 5 case series for demo	• Tune MO similarity threshold • Explainability: 3 key fields • Cross-district suspect match

Jul 15 — Predictive Intelligence

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Trend chart (Chart.js) • Early warning badge on map • PDF export button	• LLM predictive questions • Explanation with source records • Wire Prophet into chat answers	• Seed seasonal pattern data • Verify trend date range queries • Index on date + location cols	• Prophet per-crime-type forecast • Early warning threshold logic • Top 3 predicted hotspots

Jul 16 — Full Flow Test + Fixes

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• End-to-end UI test all features • Fix display/render issues • PDF export with history working	• Full API integration test • Fix LangChain context bugs • Latency check: target <3s	• DB integrity check all tables • Load test: 10 concurrent queries • Backup Supabase before demo	• ML accuracy check all 3 models • Fix prediction edge cases • Document confidence meanings

Phase 4 · Polish & Demo (Jul 17–19)

Jul 17 — Demo Script + Dry Run 1

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Polish UI colours & spacing • Record 3-min demo screen flow • Fix any mobile layout issues	• Write demo script (exact queries) • Dry run 1: full team walkthrough • Prepare judge Q&A; doc	• Prepare 'wow' demo data • Keep-alive ping for Supabase • Ensure Neo4j graph is rich	• Prepare ML insight visuals • Verify model outputs in UI • Dry run: cross-case match live

Jul 18 — Pitch Deck + Dry Run 2

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Final UI polish pass • Screenshots for pitch deck • Test on multiple browsers	• Build pitch deck (full slides) • Dry run 2: timed 5+3 min • Prepare 1-page summary	• Final evidence hashes check • Verify all photos load in demo • Document schema for submission	• Explainability panel demo moment • Write ML approach for deck • Full early warning alert test

Jul 19 — Final Day — Submit & Present

Gopika S	Abishek K S	Dharaneesh J	Vishal K
Full Stack · Frontend	Full Stack · PM	Database Manager	ML Engineer
• Production build React app • Deploy to Vercel / Netlify • Final UI smoke test live URL	• Deploy FastAPI to Railway/Render • Final API smoke test production • Submit project & deck	• Final Supabase + Neo4j check • Keep DB active during pitch • Ready for DB judge questions	• Final ML test on production • Backup static ML outputs ready • Present ML section in pitch